Jyothis Chandran

Final-year PhD candidate · CMB component separation

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Research profile

I work on component separation for CMB experiments, using needlet-based blind and semi-blind methods: NILC, GNILC, constrained ILC and multi-Stokes Hybrid NILC. I apply these to reconstruct the thermal Sunyaev–Zeldovich Compton-y signal and to suppress Galactic and extragalactic foregrounds. I am lead author of the current *Planck* PR4 NILC y-map and of its polarization-aided, CIB-deprojected variant. Both are public, through Zenodo and the *Planck* Legacy Archive, and are used by the CMB and LSS communities. My current work extends these methods to the extraction of challenging Galactic foregrounds such as free-free emission, and to machine-learning approaches for point-source detection in CMB data. I am also extending this work to cross-correlations of component-separated maps with galaxy surveys, to trace the redshift history of hot gas and dust.

Education

Jun 2022 – exp. Mar 2027
Santander, Spain

PhD in Science and Technology (Astrophysics and Particle Physics)
Universidad de Cantabria / Instituto de Física de Cantabria (IFCA)
Thesis: *Teasing new cosmological observables out of future CMB and radio surveys*
Supervisors: Dr R. Belén Barreiro and Dr Mathieu Remazeilles
Expected submission: 28 March 2027
Developing component-separation methods for faint cosmological signals in CMB and radio data, and delivering new algorithms and all-sky maps to the community.

2020 – 2021
Vitória, Brazil

Provisional enrolment, PPGCOSMO doctoral programme in Cosmology
Universidade Federal do Espírito Santo (UFES)
Admitted to a joint Brazil-Germany doctoral project under Dr Oliver F. Piattella (UFES), co-supervised by Prof. Luca Amendola (Heidelberg University), and completed graduate coursework remotely. Enrolment could not be taken up, owing to pandemic-related visa delays and a subsequent change in supervisory arrangements.

2015 – 2020
Thiruvananthapuram, India

BS-MS Dual Degree in Physics (Physics major, Mathematics minor)
Indian Institute of Science Education and Research (IISER-TVM)
CGPA 8.92/10. Thesis: *Harmonic-domain Internal Linear Combination for CMB analysis and the effect of biases* (Dr Soumen Basak).

Publications

2 first-author refereed papers · 2 conference proceedings · 4 LiteBIRD collaboration papers

First-author refereed papers

P1 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization and deprojection.* JCAP **08**, 062 (2026). doi:10.1088/1475-7516/2026/08/062, arXiv:2603.17799.
Introduced a multi-Stokes Hybrid NILC that uses Planck Q/U channels as auxiliary inputs with zero tSZ response, together with selective deprojection of CIB spectral moments. I adapted the method, ran the simulations and the PR4 analysis, and wrote the paper.

P2 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *An improved Compton parameter map of the thermal Sunyaev–Zeldovich effect from Planck PR4 data.* MNRAS **526**, 5682–5698 (2023). doi:10.1093/mnras/stad3156.
Updated tSZ Compton-y map from Planck PR4 (NPIPE) data. I modified the NILC pipeline, carried out the statistical analysis and produced the public map. I developed the work and wrote the paper.

Conference proceedings

C1 **J. Chandran**. *Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization.* Proceedings of the 60th Rencontres de Moriond (Cosmology, 2026), accepted.

C2 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *An updated and improved thermal SZ y-map from Planck PR4 data.* EPJ Web Conf. **293**, 00010 (2024), *mm Universe 2023*.

Other papers (LiteBIRD collaboration)

L1 K. Aizawa *et al. LiteBIRD mission overview after mission reformation.* arXiv:2607.25285 (2026).

L2 S. Vinzl *et al. Galactic science with the LiteBIRD satellite: spectral characterization of diffuse Galactic polarized emission at the angular power spectrum level.* arXiv:2607.20080 (2026).

L3 A. Novelli *et al. A cross-calibration approach for polarisation-sensitive detectors in CMB experiments: application to LiteBIRD's polarisation angle calibration.* arXiv:2606.23207 (2026).

L4 N. Raffuzzi *et al. Mitigating half-wave plate systematics at the map-making level: calibration requirements for LiteBIRD.* arXiv:2602.02410 (2026).

Public data releases

D1 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *CIB-deprojected Compton y-parameter map of the thermal SZ effect from Planck PR4 data.* Zenodo (2026). doi:10.5281/zenodo.18876142
All-sky Hybrid NILC y-maps with selective CIB moment deprojection. 110 downloads since March 2026 (Zenodo, Oct 2026). Used for FRB–tSZ cross-correlations (Zhai et al., arXiv:2608.09014).

D2 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *Compton y-parameter map of the thermal SZ effect from Planck PR4 data.*
Released through the **ESA *Planck* Legacy Archive**, ESA's official repository for *Planck* data products, and through Zenodo (2023), doi:10.5281/zenodo.7940376
Released with the masks used. Used by other groups, e.g., for the hot gas around the Sombrero galaxy (Li et al., ApJ 1009, 72, 2026) and for FRB–tSZ cross-correlations (Zhai et al., arXiv:2608.09014). 655 downloads and 1,166 record views (Zenodo, Oct 2026).

Doctoral research

2022 – 2023
Thermal SZ Compton-y reconstruction from Planck PR4
Implemented a harmonic ILC pipeline and modified the needlet ILC pipeline in Python, applying both to Planck PR4 (NPIPE) simulations and data and selecting NILC after a like-for-like comparison. Deconvolved the measured PR4 channel beams rather than assuming Gaussian beams, and produced HR1/HR2 half-ring maps to characterise residual noise. Carried out the statistical analysis: one-point statistics, pseudo-C_ℓ spectra with NaMaster, and skewness. The resulting map has 7% lower thermal noise and 34% less CIB contamination than the public PR2 y-map, and yields $\sigma_8 = 0.76 \pm 0.02$ from map skewness, indicating that the PR2 σ_8 tension was not driven by systematics. Also built a new point-source mask from the per-channel PR4 catalogues.

2024 – 2026
Multi-Stokes Hybrid NILC and CIB moment deprojection
Extended NILC to take Planck Q/U maps as auxiliary channels constrained to zero tSZ response, exploiting the intrinsic temperature–polarization correlation of Galactic dust and synchrotron while preserving the unpolarized tSZ signal. Added constrained deprojection of the CIB SED and its first-order temperature moment over the last five needlet bands, trading CIB suppression against the noise penalty from the enlarged constraint space. On PR4 the Hybrid TQU map shows 40–45% lower cross-correlation with the IRAS 100 μm dust tracer than the temperature-only map; at LiteBIRD-like sensitivity, simulations show a further $\sim$20% reduction in projected Galactic residual power.

2026 – present
Non-parametric reconstruction of Galactic free-free emission
Building and validating ILC and NILC pipelines to extract full-sky free-free emission without parametric templates, systematically exploring frequency selection, masking, localisation and deprojection schemes against residual dust, synchrotron and anomalous microwave emission. Applying the optimised configuration to Planck PR4 and combining with complementary low-frequency surveys.

2026 – present
Machine-learning point-source detection
Working on convolutional neural network methods for detecting extragalactic point sources in CMB total-intensity maps, trained on simulations, within RadioForegroundsPlus.

2024 – present

Towards a LiteBIRD forecast for tSZ × ISW
Early-stage work within the LiteBIRD thermal SZ group: quantifying contaminant-induced biases in tSZ-deprojected CMB maps and CMB-deprojected tSZ maps, and comparing simulated tSZ × CMB cross-spectra with theoretical predictions, to estimate the limit LiteBIRD could set on the tSZ × ISW cross-correlation.

Research projects & funding

2022 – present

Spanish national research projects, MCIN / AEI
Research carried out within PID2022-139223OB-C21, PID2022-140670NA-I00 and PID2025-173927NB-I00, all three of which have supported my conference travel. Additional support through the IFCA structural research programme of the Gobierno de Cantabria.
PID2022-139223OB-C21: *Frontier research in Cosmology with LiteBIRD* (PIs: Dr E. Martínez-González, Dr P. Vielva).
PID2022-140670NA-I00: *Unveiling the Cosmos with Line-Intensity Mapping Experiments* (PI: Dr J. L. Bernal).

2024 – 2026

RadioForegroundsPlus (Horizon Europe) (HORIZON-CL4-2023-SPACE-01, GA 101135036)
Coordinator: Dr J. A. Rubiño-Martín (IAC); IFCA node led by Dr R. Belén Barreiro, with Dr Mathieu Remazeilles as project manager. Contributed to Galactic foreground mitigation in Compton-y maps using polarization as an external tracer (multi-Stokes Hybrid NILC; JCAP 2026); to full-sky free-free reconstruction with blind and semi-blind ILC methods; and to CNN-based point-source detection in total intensity.

2022 – 2027

"Concepción Arenal" Predoctoral Contract
Competitive doctoral contract awarded by the Universidad de Cantabria.

2015 – 2020

INSPIRE Fellowship, Ministry of Science and Technology, India
National merit fellowship for study and research in the basic sciences.

Dec 2018

CSIR-UGC NET (Physical Sciences): All India Rank 20

2018

IUCAA Visiting Students Programme award
Inter-University Centre for Astronomy and Astrophysics, Pune.

Talks & posters

Oct 2026
Tenerife, Spain

Hybrid TQU Needlet ILC to mitigate Galactic foregrounds in tSZ reconstruction (contributed talk)
Galactic Science and CMB Foregrounds, Instituto de Astrofísica de Canarias.

Sep 2026
Stockholm, Sweden

Improved foreground cleaning via polarization-aided cleaning and CIB deprojection in thermal SZ Compton-y maps (talk)
Cosmoglobe 2026: Global End-to-End Analysis at All Frequencies, Nordita. Conference week of a three-week programme.

Aug 2026
Leiden, Netherlands

Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization-aided cleaning and CIB deprojection (talk)
COSMO-26: International Conference on Particle Physics and Cosmology.

May 2026

Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization and deprojection (talk)
Simons Observatory Foregrounds Working Group meeting.

Mar 2026
La Thuile, Italy

Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization (poster; proceedings accepted)
60th Rencontres de Moriond, Cosmology 2026.

Nov 2023
Tenerife, Spain

A refined Planck thermal SZ Compton y-parameter all-sky map to probe the entire hot gas in the Universe (talk)
IX Meeting on Fundamental Cosmology, Instituto de Astrofísica de Canarias.

Jun 2023
Grenoble, France

An updated and improved thermal SZ y-map from Planck NPIPE data (talk)
mm Universe 2023: Observing the Universe at millimetre wavelengths, LPSC.

Collaborations

Jan 2024 – present

Member, LiteBIRD Collaboration
Thermal SZ Project Study Group (project: *Mapping the hot gas in the Universe*), led by M. Remazeilles and G. Luzzi. Working towards a LiteBIRD forecast for the tSZ × ISW cross-correlation.

Earlier research experience

2020 – 2021

Harmonic ILC for multi-component reconstruction
With Dr Soumen Basak (IISER-TVM) and Dr Jacques Delabrouille (APC, Paris). Extended harmonic ILC towards the simultaneous reconstruction of several components; implemented and tested the pipeline on simulated LiteBIRD skies. Methodological groundwork for the ILC techniques used in my thesis.

May – Jun 2019
Heidelberg, Germany

Visiting researcher, Institut für Theoretische Physik
Heidelberg University, with Prof. Luca Amendola and Dr Oliver F. Piattella
Tested a model-independent expression for the anisotropic stress (gravitational slip) parameter in Horndeski gravity, comparing the analytic approximations of the Bellini–Sawicki parameterisation against numerical solutions across a range of scales. Built a Python-based analysis for the hi_class output and took part in the cosmology group's weekly seminar programme.

May – Jun 2018
Pune, India

IUCAA Visiting Students Programme
With Dr Sanjit Mitra. Cross-correlation statistics of noise in laser interferometric gravitational-wave detectors.

Technical skills

Programming	Python (numpy, scipy, healpy, astropy, matplotlib, tensorflow), IDL/GDL, MATLAB, Mathematica, LaTeX, shell.
Cosmology software	HEALPix, NaMaster / pymaster, CAMB, CLASS / hi_class, Planck Sky Model (PSM), PySM.
Methods	Needlet and harmonic ILC, GNILC, constrained ILC and moment deprojection, multi-Stokes Hybrid NILC, tSZ and CIB component separation, pseudo-C_ℓ power spectrum estimation, point-source masking and machine-learning point-source detection, one-point and higher-order map statistics.
Computing	Altamira (IFCA-CSIC, Spanish Supercomputing Network) and NERSC (active account); Slurm, Git, Linux.
Languages	English (TOEFL 112/120), Malayalam (native), Hindi, Spanish.

Outreach

2019

IISER-TVM schools outreach programme
Interactive session on radiation pressure and electromagnetic waves for secondary students, Evans Higher Secondary School, Parassala, India.

References

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Research supervisor, Heidelberg (2019)

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